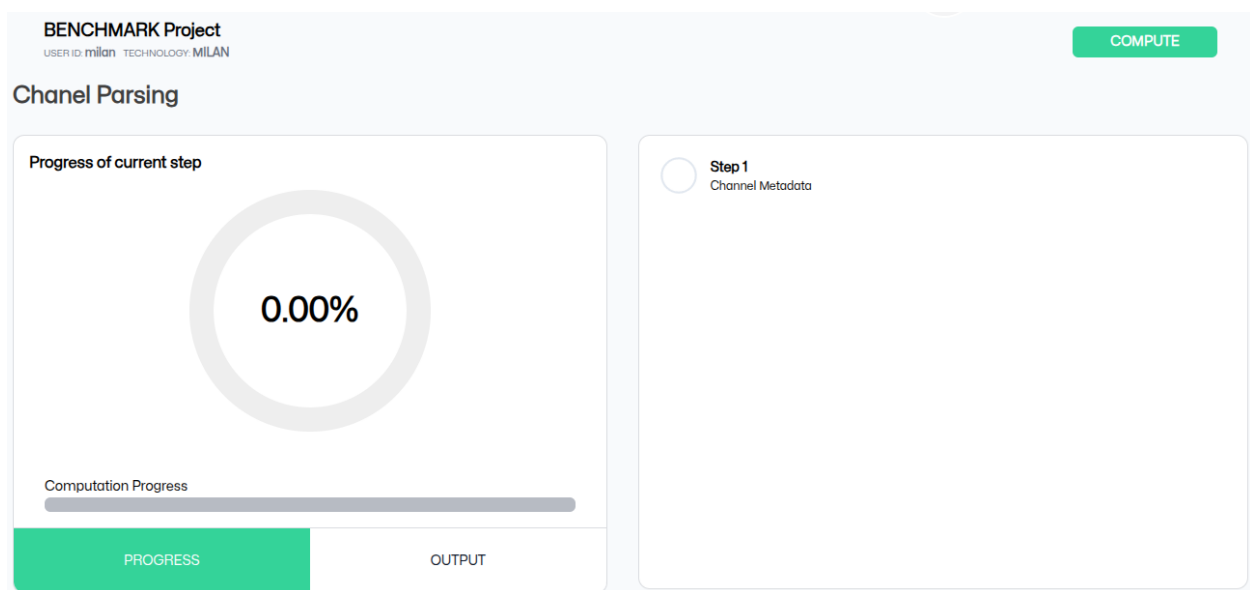


Image analysis pipeline:

The following tutorial explains how to run the analysis pipeline for DISSCOvery. The steps are outlined above.

The first step is Channel Parsing.

To first start this step, click on the ‘Compute’ button. This step will extract metadata from the czi file that was uploaded for this project.



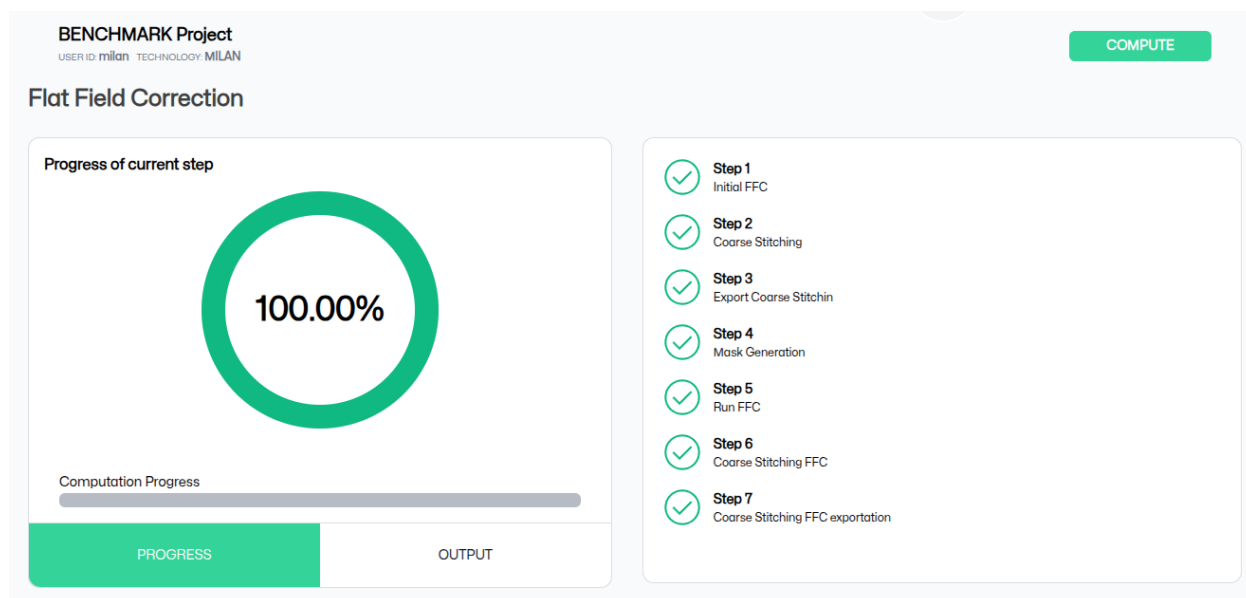
An example of the metadata file is given below. You can manually change the reference round and the reference channel, if it has not been identified correctly.

You can also add the round and marker information

Next, click on 'SAVE ACTUAL ANNOTATIONS' to save the experimental design and continue to the next step

The next step is Flat Field Correction.

To start this step, you have to click on the COMPUTE button, and the pipeline performs the following 7 steps to perform FFC.



The pipeline uses two tools to perform FFC. For DAPI channel, the FFC is performed using the raw image, the AF channel uses Kask to perform FFC and FITC, TRITC and Cy5 use BASIC to perform FFC.

After this step, the user can perform a QC of the results and can also change the algorithm if the results are not satisfactory.

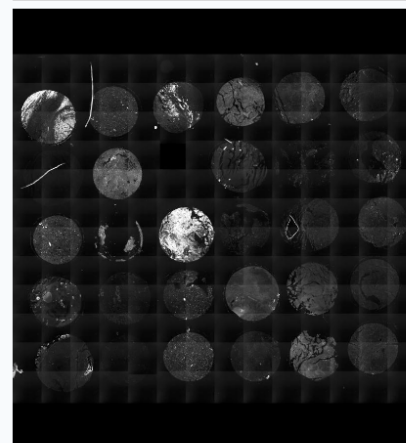
SHOW NEW METHODS

COMPUTE METHODS

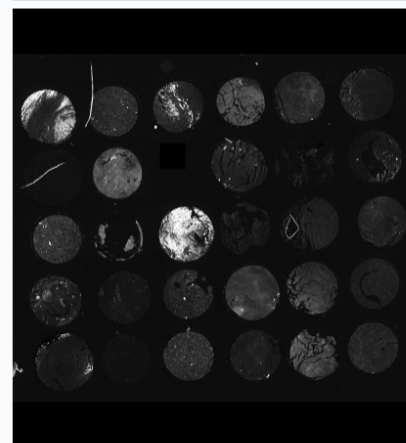
Round ↑↓	Slide ↑↓	Channel ↑↓	Version ↑↓	Method ↑↓
R00	BM	AF	V01	KASK
R00	BM	CY5	V01	BASIC
R00	BM	DAPI	V01	RAW
R00	BM	FITC	V01	BASIC
R00	BM	TRITC	V01	BASIC
R01	BM	AF	V01	KASK
R01	BM	CY5	V01	BASIC
R01	BM	DAPI	V01	RAW
R01	BM	FITC	V01	BASIC
R01	BM	TRITC	V01	BASIC
R01	BM	AF	V02	KASK
R01	BM	CY5	V02	BASIC
R01	BM	DAPI	V02	RAW
R01	BM	FITC	V02	BASIC
R01	BM	TRITC	V02	BASIC
R02	BM	AF	V01	KASK
R02	BM	CY5	V01	BASIC
R02	BM	DAPI	V01	RAW
R02	BM	FITC	V01	BASIC
R02	BM	TRITC	V01	BASIC
R02	BM	AF	V02	KASK
R02	BM	CY5	V02	BASIC
R02	BM	DAPI	V02	RAW
R02	BM	FITC	V02	BASIC
R02	BM	TRITC	V02	BASIC

CONTINUE >

BEFORE

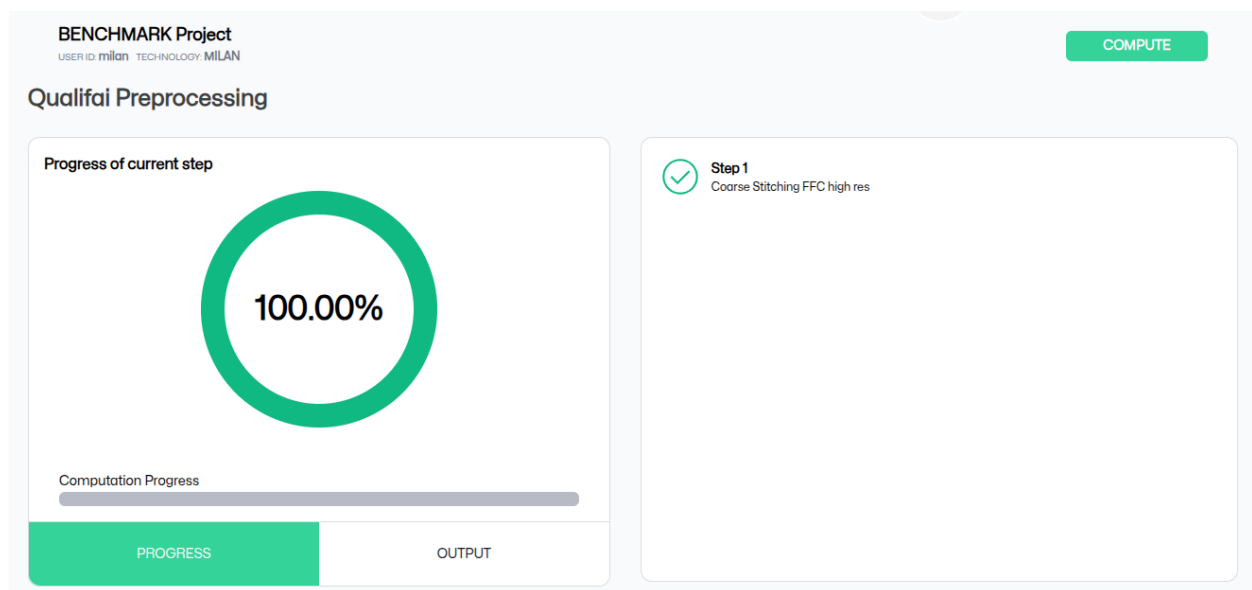


AFTER

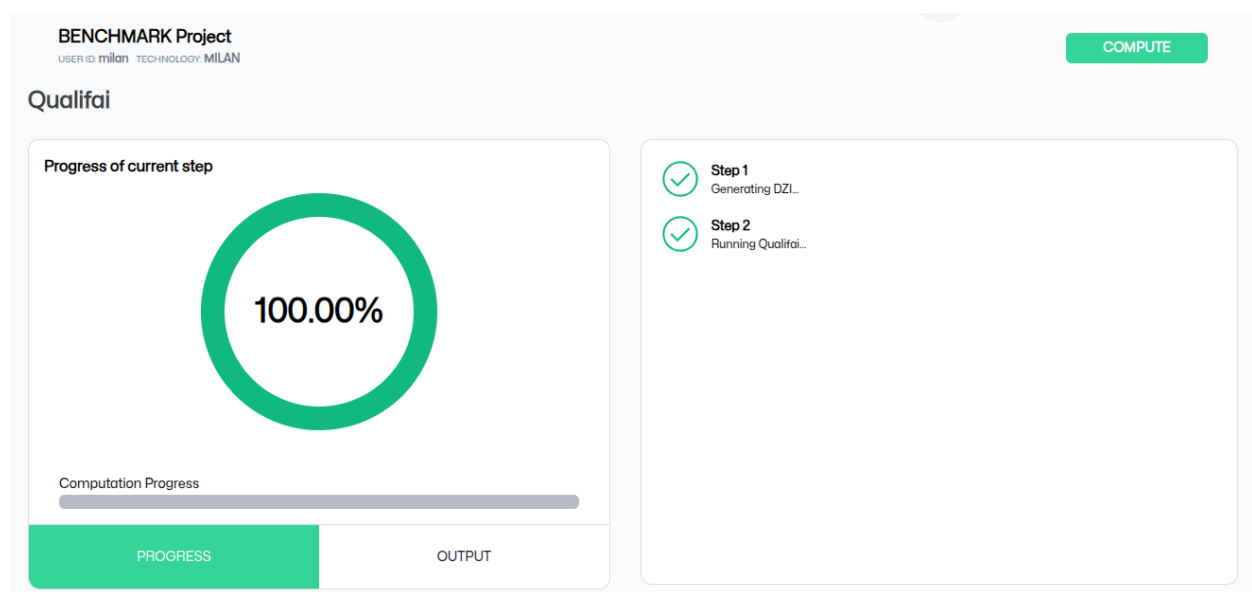


To change the tool used, you can click on **SHOW NEW METHODS**. If the results are satisfactory, click on **CONTINUE** on top right to go to the next step, which is **QUALIFAI Preprocessing**.

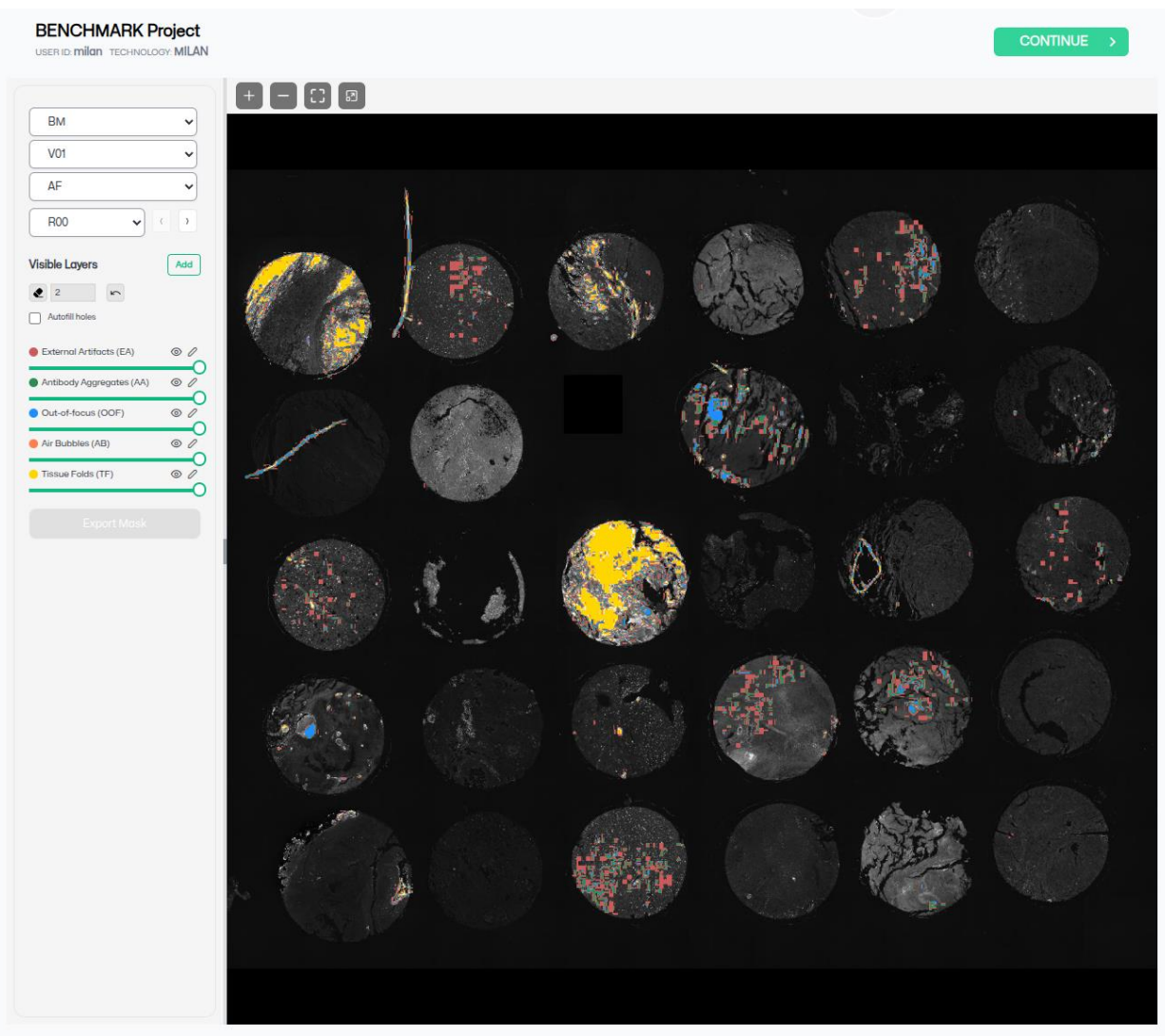
QUALIFAI is a tool that can annotate and flag out artefacts. The preprocessing step is performing a coarse stitching of the FFC output.



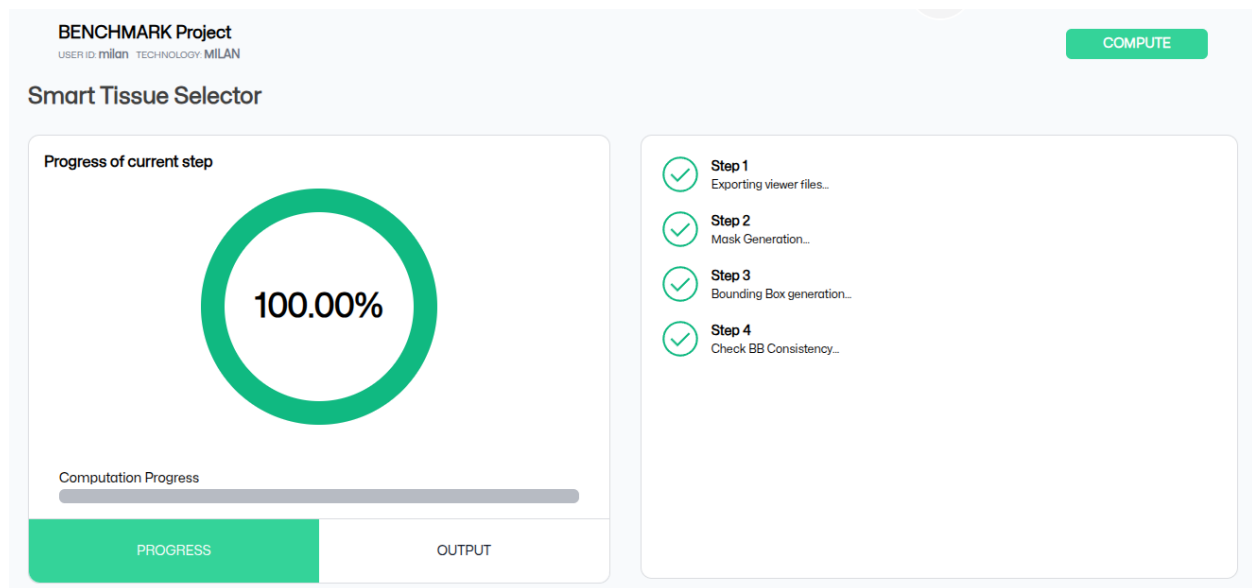
Once pre processing is performed, QUALIFAI can be run on the coarse stitched images.



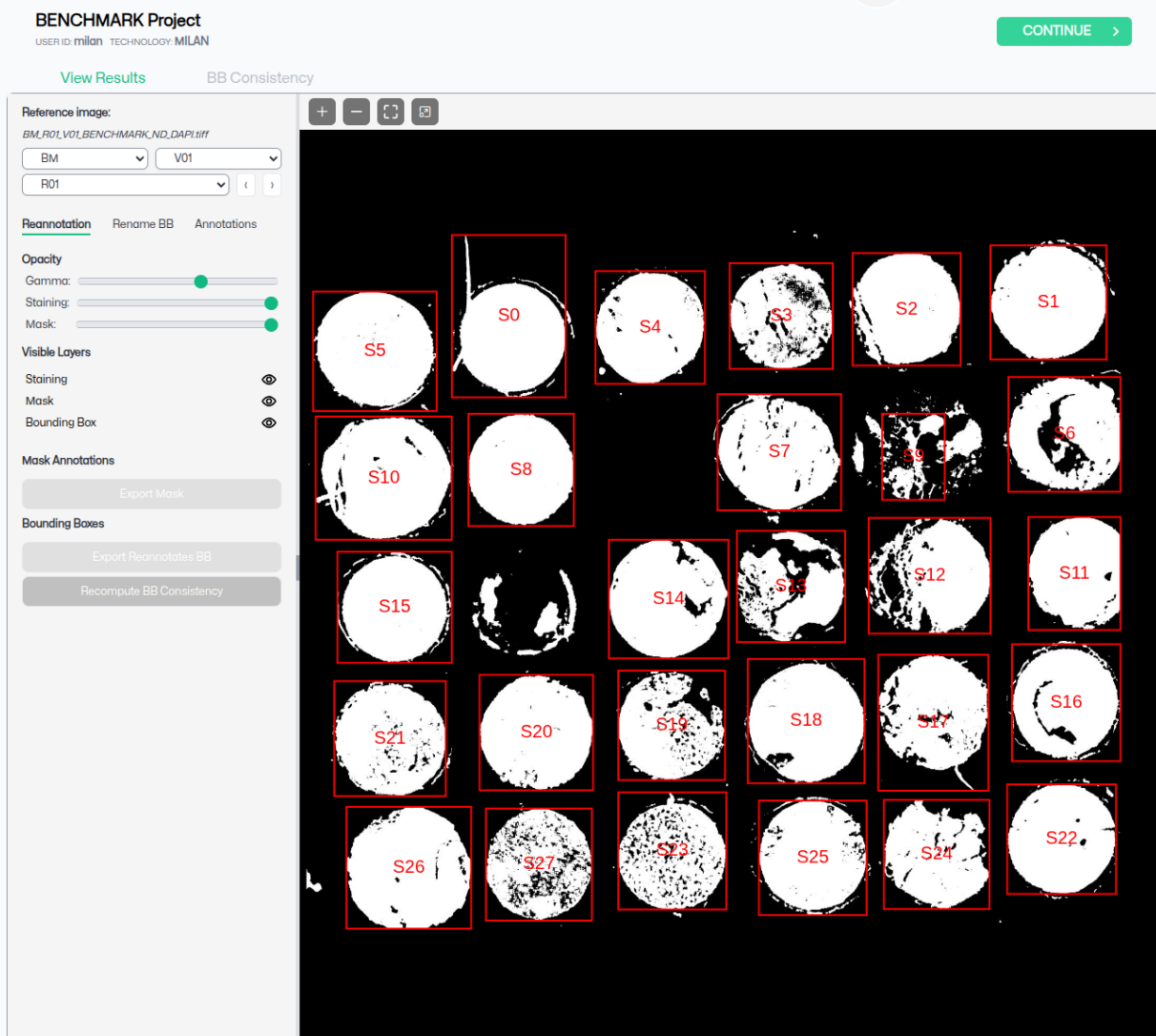
After QUALIFAI is run, the user can check the quality of the annotations, and also has the option to annotate more artefacts which may have been missed by QUALIFAI.



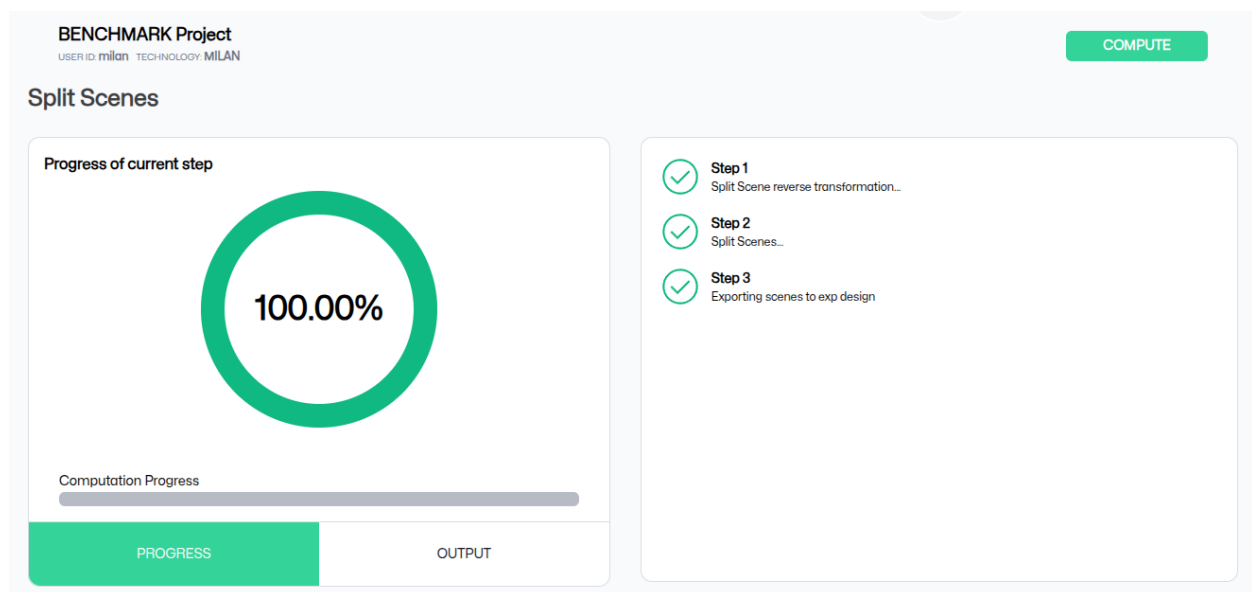
The step after is Smart Tissue Selection. In this step, the pipeline will generate bounding boxes to separate individual pieces of tissue to be analyzed.



Once this step is run, there is again the option to perform QC to make sure the correct pieces have been selected.

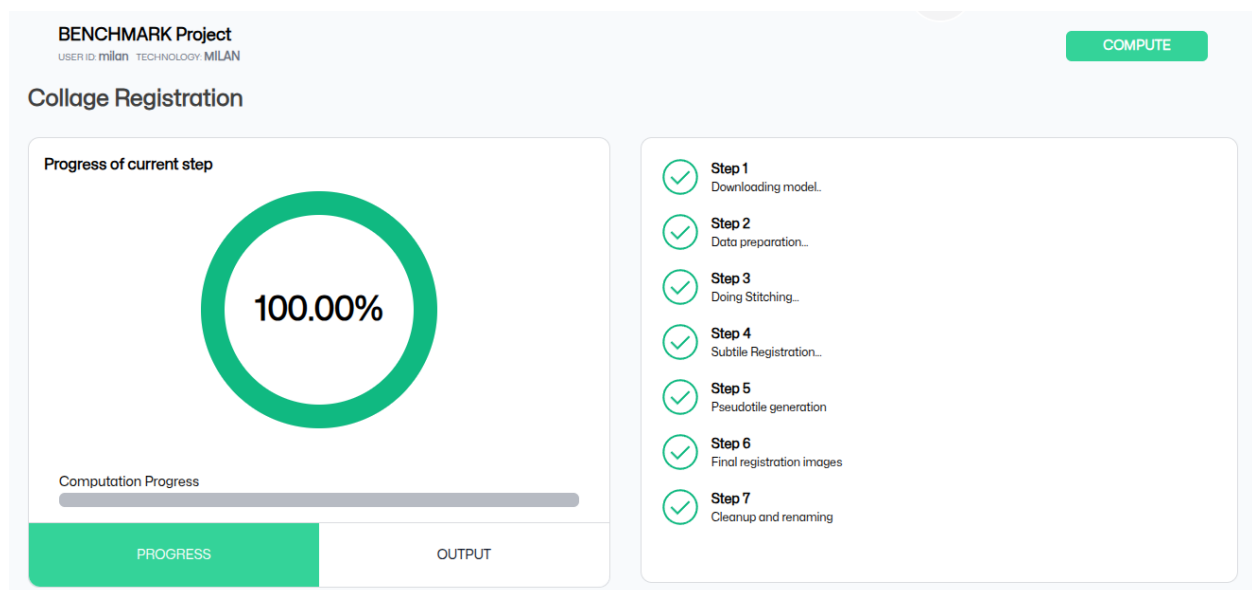


The step after STS is Split Scenes. This is where the tissue is split into different scenes for the analysis based on the output from the previous step.



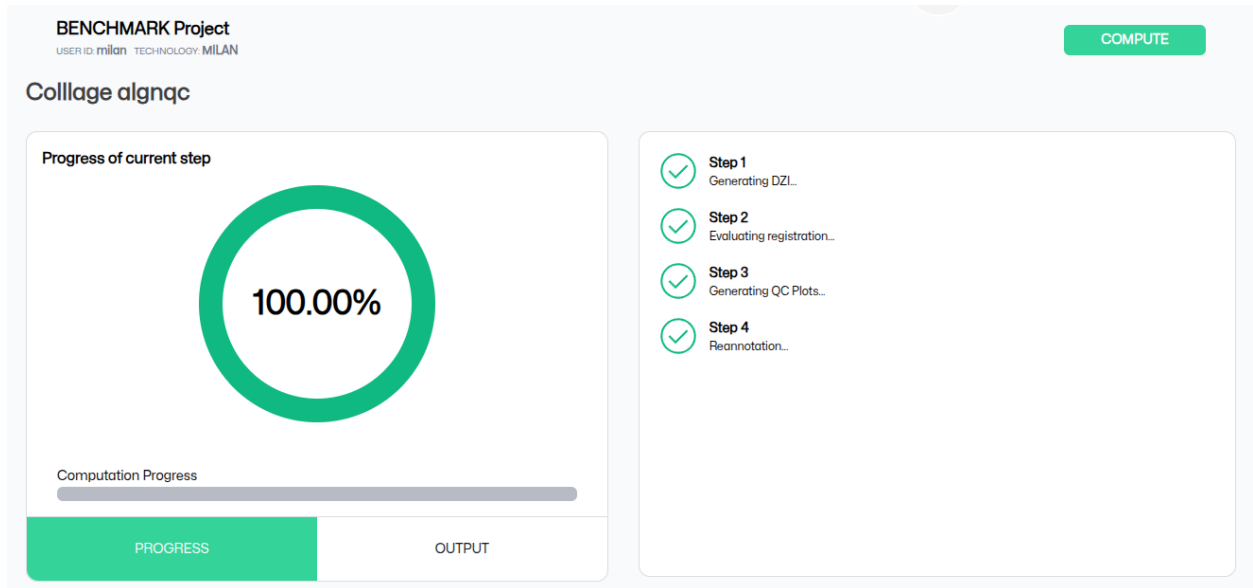
Once the scenes have been split, DISSCOvery performs registration. This is a step where images of different rounds, which are the of same scene are aligned so that the corresponding points in the two images match spatially.

Registration is performed using a tool known as COLLAGE.

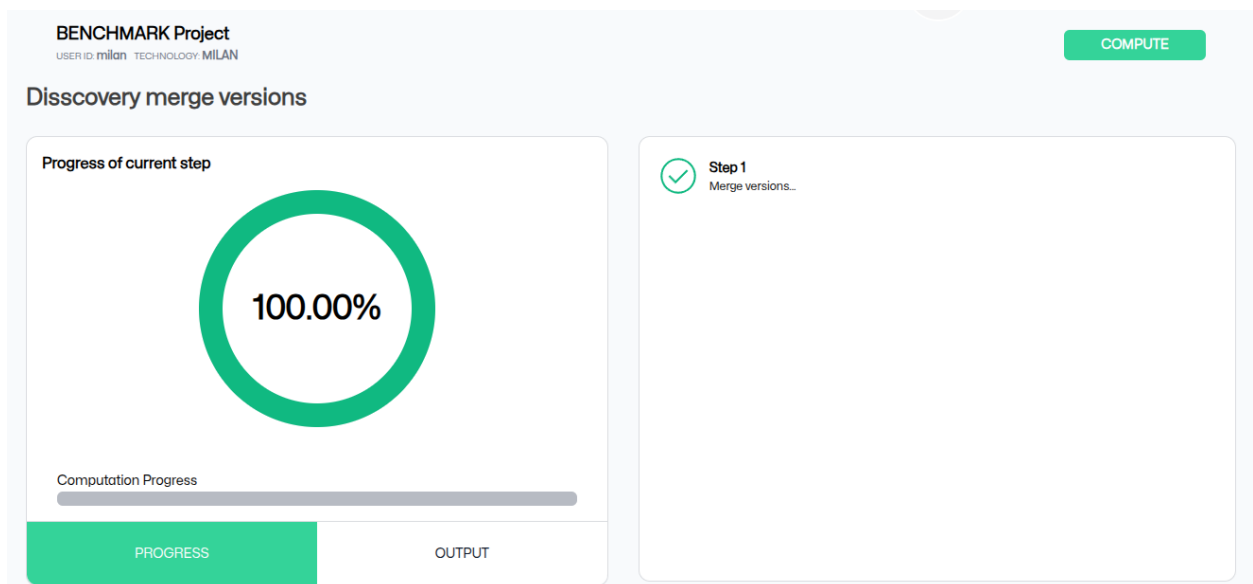


As a QC step to make sure the images were registered correctly, the next step is a tool known as AlignQC which will generate a QC image, which is an overlap of the fixed image in

R01 against images from other rounds. Regions of poor registration can also be manually annotated at this step.



NEED A GOOD EXPLANATION FOR MERGE VERSIONS



Cell Identification

The steps after Downstream Preprocessing are Normalization, Cell Identification and Digital Reconstruction.

Cell Identification is done in a hierarchical structure, where cell types can be identified and individual cell types can be further clustered into subtypes. The first step is Normalization and Sampling. Normalization of marker intensity is performed for each scene and each marker. They are in the form of a Z-score, and are in the range of -5 to $+5$.

Data sampling is also performed. In the example shown, 25000 cells are selected in total at random from each scene and clustering is performed using three

The screenshot shows a web interface titled "TEST_MILAN" with a trash icon in the top right corner. Below the title bar, there is a section with a grey background containing the following elements:

- A label "Total cells" followed by the value "502.413".
- Two checked checkboxes: "Do Normalization" and "Do Sampling".
- A label "Cells:" followed by a numeric input field containing "25000", with minus and plus buttons on either side.
- A section titled "Versions available" with an empty white box below it.
- A dropdown menu at the bottom with the text "Select an option" and a downward arrow.

